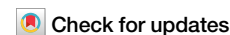


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Beyond the K-valley: exploring unique trion states in indirect band gap monolayer WSe₂



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We calculate the optical properties of monolayer WSe₂ using ab initio many-body screened configuration interaction. For this indirect band gap material we describe additional species of negatively charged trions involving the **Q**-valley which are energetically more favorable than those formed at the **K**-valley. While the results are in excellent agreement with experimental data, they shed new light on observed but yet unaddressed optical features. The analysis of the many-body interactions reveals mechanisms behind increased singlet-triplet splitting and redshifted energies in these **Q**-valley trions, compared to **K**-valley trions.

Extensive research is currently focused on two-dimensional (2D) systems due to their favorable physical characteristics, including substantial band gaps, large spin-orbit coupling (SOC) effects, and strong light-matter interaction^{1–4}. Additionally, the tunability of electronic and optical properties via external fields, strain, stacking, and twisting makes 2D materials an interesting playground for theoretical and experimental studies^{5–9}. The reduced dimensionality of these systems results in unconventional screening of the Coulomb interaction, thereby generating strongly bound neutral and charged excitons^{10–13}. One of the most renowned groups of 2D materials are layered transition metal dichalcogenides (TMDs) with significant potential for future applications^{14,15}. TMDs can be represented as MX₂ where M = (Mo, W) and X = (S, Se, Te) and crystallize in the hexagonal 2H-phase crystal phase. Upon exfoliation to a single layer, most TMDs undergo an indirect-to-direct semiconductor transition, resulting in a direct optical gap at the **K**-valley^{16–18}. Thus, many previous experimental and theoretical studies on optically excited states in monolayer TMDs rely on explanations based solely on the **K**-valley to account for the prominent spectral features of absorption and photoluminescence (PL)^{19–27}. While this assumption may hold true for most TMDs, compelling data, both experimental and theoretical, indicates the presence of an indirect band gap for atomically-thin WSe₂^{28–33}. Whereas the valence band maximum (VBM) remains situated at **K**, the conduction band minimum (CBM) is found at the **Q**-valley, halfway along the Γ -K direction^{29,31,32} (see Fig. 1). As such, an additional electron would preferentially localize at **Q** rather than **K** for low temperatures, which impacts the compositions and energies of low-energy negative trions. Absorption and PL spectra on WSe₂ reveal a distinctive two-peak structure for negatively-charged

trions^{21,22,24–26,34–40}. The prevailing belief is that these originate from configurations of electrons and holes located near **K/K'** (denoted X_K[−]).

We show that monolayer WSe₂ hosts negative trion species with a total center-of-mass momentum **Q**_{COM} = **Q** (denoted X_Q[−]), where one carrier is located at **Q**, as recently suggested by Perea-Causin et al.⁴¹ based on effective mass (eff. mass) calculations. We find that X_Q[−] and X_K[−] produce similar absorption features, with X_Q[−] energies slightly red-shifted relative to those of X_K[−]. Additionally, we find singlet-triplet ordering for both trion species and reveal the relevant mechanisms for the trion fine-structure splitting beyond independent-particle (IP) effects. Our insights into WSe₂ trion species not only impact the fundamental understanding of excitonic physics but also bear practical applications in optoelectronics. Specifically, the emerging field of valleytronics, where one seeks to exploit the electron's valley degree of freedom (e.g., in valley-based transistors and switches, where information can be encoded in the valley degree of freedom, photodetectors and light-emitting devices that leverage valley polarization, or even quantum computing components, where valleys can serve as qubits), may profit from this type of excitation.

Results

Figure 1a shows the DFT-relaxed crystal structure of monolayer WSe₂, and the corresponding electronic structure is depicted in Fig. 1c, where we find an energetic offset in the lowest conduction band between **Q** and **K** of 82 meV. The SOC induced spin-splitting in the valence band is computed to be ~455 meV⁴², an order of magnitude larger than the SOC-induced conduction band splitting (CBS) of 44 meV. Our calculations overestimate the CBS at **K**, which is experimentally observed to be 10–15 meV^{27,43}, prompting us to manually adjust the splitting to 10 meV.

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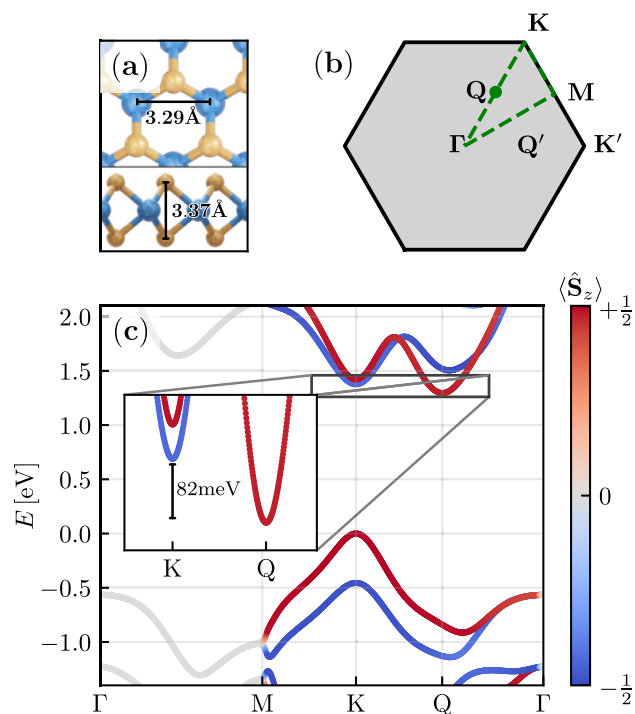


Fig. 1 | Unit cell and bandstructure of monolayer WSe₂. **a** WSe₂ in top and side view with the in-plane lattice constant and Se – Se distance. **b** The first Brillouin zone with the prominent high-symmetry points and corresponding **k**-path (green) as used in **c** the bandstructure computed within DFT. The color indicates the spin expectation value along **z**. The inset shows the energy difference of the conduction band at **K** and **Q**.

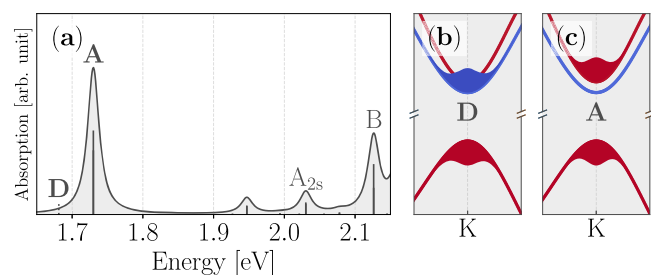


Fig. 2 | Low-energy excitonic states. **a** Absorption spectrum for **X** using a phenomenological broadening of 10 meV. **b**, **c** Configurational weights displayed on the bands near **K** of states **D** and **A**, respectively.

Excitons

In Fig. 2a the absorption spectrum of excitons (**X**, $Q_{\text{COM}} = 0$) is displayed, where the entire spectrum is shifted so that the first bright exciton, **A** (see Fig. 2c), is located at the experimental value of 1.73 eV^{21,24–26,34,39}. Three optically-active bound excitons, denoted as **A**, **A_{2s}**, and **B**⁴⁴, can be identified, with a predicted binding energy of 480 meV for the **A** exciton. We further observe a bright state close to the **A_{2s}** transition (see Supplementary Information (SI)). All reported values are found by extrapolating to infinitely dense **k**-samplings in order to ensure high accuracy. The splitting between the **A** and **B** excitons (here 388 meV) arises mainly due to SOC splittings of the valence and conduction bands, consistent with reported experimental values (380 meV³⁹). Furthermore, we indicate the lowest exciton in the spectrum with **D**, which is a spin-dark state located 49 meV below the **A** exciton as seen in Fig. 2b. We isolate the dominant contributor to the **D**-**A** splitting to be the coupling between configurations, making it almost five times larger than the CBS.

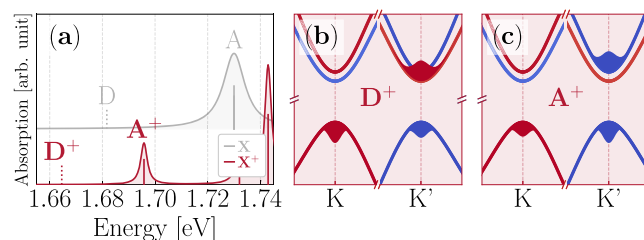


Fig. 3 | Low-energy positively charged trions centered around **K.** **a** Absorption spectrum for **X⁺** and **X** (offset along the **y**-axis) in red and grey, respectively. **A** phenomenological broadening of 1.5 meV has been used for the trions. **b**, **c** Configurational weights near **K** and **K'** for states **D⁺** and **A⁺**.

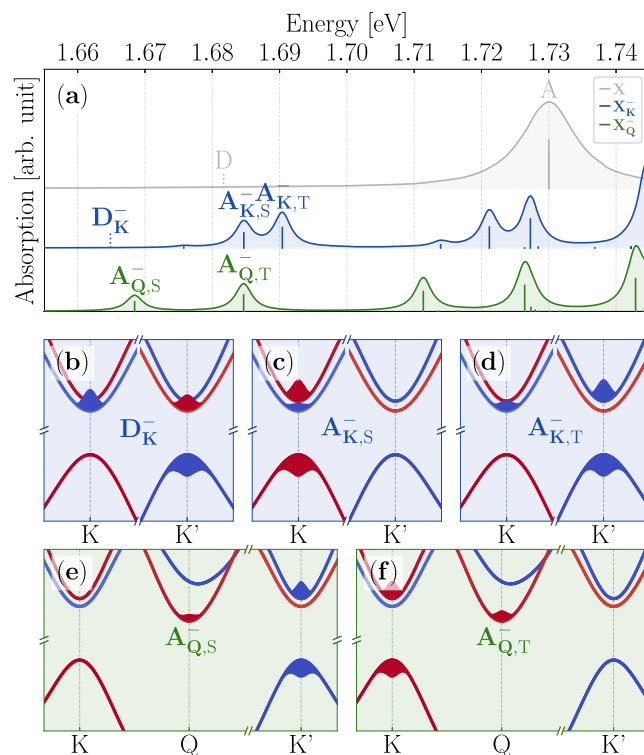


Fig. 4 | The two species of low-energy negatively charged trions. **a** Absorption spectrum for **X** (grey), **X_K⁻** (blue) and **X_Q⁻** (green). The spectra are shifted for better comparison. A phenomenological broadening of 1.5 meV has been used for the trions. Configurational weights for **X_K⁻** (**b**–**d**) and **X_Q⁻** (**e**, **f**), where the weight on **Q** is scaled by 1.5 for better visibility.

Positively charged trions

For positively charged trions (**X⁺**) the spectrum in Fig. 3a shows only a single bright resonance, **A⁺**, with a binding energy of 31 meV relative to **A**, as the VBM splitting is comparably large. **A⁺** is accompanied by a satellite dark state, **D⁺**, with a dark-bright splitting of 33 meV. As shown in Fig. 3b, c, the positive trions resemble either **D** or **A** excitons formed at **K'** plus an additional hole at **K**. Compared to the excitons, the configurational weights are more localized in reciprocal space.

Negatively charged trions

Figure 4a displays the absorption spectra for **X_K⁻** in blue and **X_Q⁻** in green. For **X_K⁻** we identify two bright trion states, **A_{K,S}⁻** and **A_{K,T}⁻**, which exhibit binding energies of 43 meV and 37 meV with respect to **A**. Figure 4c, d reveals that these states resemble a more localized version of the **A** exciton in reciprocal space with an additional carrier at **K**. **A_{K,S}⁻** is an intra-valley singlet state, with all configurational weights localized near **K**. In contrast, **A_{K,S}⁻** is an intervalley triplet, where the eh pair localizes at **K'**. In our calculations **A_{K,S}⁻**

Table 1 | Comparison of binding energies for excitons and trions from both experiment (Exp.) and theory (Theo.).

	D ⁺	D ⁻	D	A _{Q,S} ⁻	A _{Q,T} ⁻	A _{K,S} ⁻	A _{K,T} ⁻	A ⁺
²⁷ Exp.	–	15	41	48 ^a	–	35	29	–
⁴⁵ Exp.	15	16	39	47 ^a	–	34	27	20
²⁴ Exp.	15	16	42	–	–	37	30	24
²⁵ Exp.	16	14	41	–	–	35	29	21
²⁶ Exp.	–	–	–	–	–	35	29	–
²⁰ GW-BSE	15	15	80	–	–	62	65	30
⁴¹ Eff. mass	12	12	57	46	–	24	–	11
This work	15	15	49	61	44	43	37	31

The binding energies of bright states (A_i⁺) are relative to A, whereas the binding energy of dark states (D_i⁺) are given with respect to D. All values in units of meV.

^aHypothesized peak origin.

appears slightly less optically-active than A_{K,T}⁻, because A_{K,T}⁻ has greater configurational weight in the first conduction band at K. Furthermore, D_K⁻ appears dark for the same reasons as D and D⁺ as shown in Fig. 4b. It is located ~15 meV below D and 64 meV below A. We emphasize, that both splittings and energetic orderings are in excellent agreement with experimental studies for X_K^{-24–27,35}, see Table 1.

So far, results have been limited to excited states localized near K/K', but Fig. 4a indicates that additional trion species, X_Q⁻, appear in the same energetic window.

Similar to X_K⁻, we unravel two distinct bright trion species, denoted as A_{Q,S}⁻ and A_{Q,T}⁻, with binding energies of 61 meV and 44 meV relative to A, redshifted with respect to X_K⁻, in agreement with ref. 41.

The composition and energetic ordering of these states is analogous to those of X_K⁻, albeit with a carrier localized at Q instead of K, which can be seen in Fig. 4e, f. They appear less bright in Fig. 4a since the configurational weight on the dopant electron at Q is less pronounced.

Indeed, our findings reveal agreement with experimental results within a few meV for all states of interest displayed in Table 1. Interestingly, Ren et al.²⁷ and Jindal et al.⁴⁵ both report an additional peak ~13 meV below the A_{K,S}⁻ state. We suggest that this additional peak aligns with our predicted A_{Q,S}⁻ state found 17 meV below A_{K,S}⁻. Moreover, the recombination of the A_{K,S}⁻ and A_{Q,T}⁻ states within the same valley results in light emission polarized in the same direction, posing a challenge for experimental differentiation due to their similar polarization.

Due to the multi-valley composition of X_Q⁻, the classification in intra- and inter-valley trions is not applicable to A_{Q,S}⁻ and A_{Q,T}⁻, as opposed to A_{K,S}⁻ and A_{K,T}⁻. Instead, the spin character of the constituting carriers hints at what determines the energetic ordering.

Discussion

Despite the similarities between X_K⁻ and X_Q⁻, our results highlight two major differences; (i) X_Q⁻ are more strongly bound and (ii) the singlet-triplet splitting (STS) for X_Q⁻ is larger than that of X_K⁻. We attribute the lower energies of X_Q⁻ partially to the slightly larger effective mass of the lowest conduction band at Q compared to K - in agreement with ref. 41. Artificially adjusting the effective mass of the lowest conduction band at K to equal that of Q redshifts the bright states of X_K⁻ by 2 meV. The remaining energy difference is likely due to variations in the coupling strength between the configurations of each species.

To understand the magnitude of the STS, we examine the contributions from the individual terms of the Hamiltonian, illustrated in Fig. 5. At the IP level there is no STS for either trion species. The introduction of diagonal interactions instigates an STS, and the Hamiltonian has four contributions (See SI); the IP energies, eh interaction between the A-like eh pair (black arrows in Fig. 5), eh interaction between the hole and the dopant electron (K^{eh}), and electron-electron (ee) interaction between the two

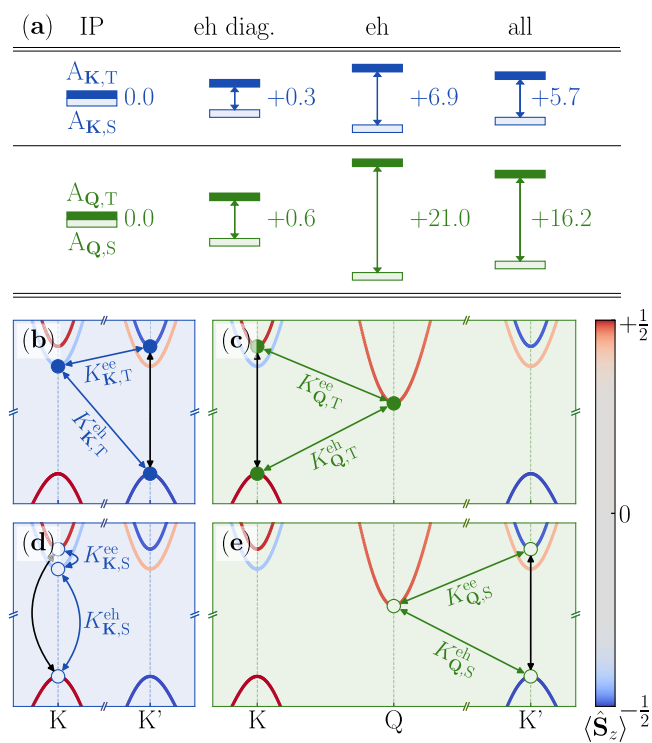


Fig. 5 | Schematic of interactions relevant to the singlet-triplet splitting of negatively charged trions. a Splitting of the singlet (light) and triplet (dark) states for X_K⁻ (blue) and X_Q⁻ (green) in meV including successively more interaction terms: only the IP energies, adding only diagonal eh interaction (eh diag.), adding off-diagonal eh interactions (eh), and finally adding electron-electron interactions (all). **b–e** Relevant eh exchange coupling elements for the singlet-triplet splitting of X_K⁻ and X_Q⁻. The black arrow indicates the A-like interaction shared by all four configurations. In each figure, the spin expectation value is encoded in the color of the bands.

electrons (K^{ee}), giving rise to an energy of

$$E_{\lambda} = \epsilon_{\lambda} + (K_{\lambda}^{\text{eh}} - J_{\lambda}^{\text{eh}}) + (K_{\lambda}^{\text{eh}} - J_{\lambda}^{\text{eh}}) - (K_{\lambda}^{\text{ee}} - J_{\lambda}^{\text{ee}}),$$

where λ is a trion state index and ϵ_{λ} the corresponding sum of trion state IP energies. Furthermore, J indicate direct Coulomb matrix elements, whereas K denote exchange matrix elements (not to be confused with the reciprocal K-point). J_{λ}^{eh} and K_{λ}^{eh} represent the A-like eh interaction between the VBM and the second conduction band at K/K', which are terms shared by all configurations of interest, marked as black arrows in Fig. 5b–d, f. For X_K⁻, the STS is then

$$E_{\text{K,T}} - E_{\text{K,S}} = (K_{\text{K,T}}^{\text{eh}} - K_{\text{K,S}}^{\text{eh}}) - (K_{\text{K,T}}^{\text{ee}} - K_{\text{K,S}}^{\text{ee}}),$$

revealing that only eh and ee exchange interactions are responsible for the STS, which analogously holds true for X_Q⁻. Strong eh exchange interactions occur between states of equal spin. However, the lowermost conduction band at Q is more spin-polarized than at K. Therefore, $K_{\text{Q,T}}^{\text{eh}}$ is larger in magnitude than $K_{\text{K,T}}^{\text{eh}}$, while the opposite holds true for $K_{\text{Q,S}}^{\text{eh}}$ and $K_{\text{K,S}}^{\text{eh}}$. In other words,

$$\underbrace{K_{\text{Q,T}}^{\text{eh}} - K_{\text{Q,S}}^{\text{eh}}}_{0.6\text{meV}} > \underbrace{K_{\text{K,T}}^{\text{eh}} - K_{\text{K,S}}^{\text{eh}}}_{0.3\text{meV}}.$$

Since the true trion states are superpositions of configurations near K, K' and Q, we argue that tendencies observed on the diagonal carry over when allowing for coupling between configurations introduced by off-diagonal

matrix elements. Fig. 5a reveals that allowing for off-diagonal eh interactions magnifies the splittings, while in contrast, the repellent ee exchange interactions decrease the STS, albeit less pronounced due to the screened ee kernel.

Lastly, the IP energy difference between the lowest conduction band at **K** and **Q** is highly sensitive to and as such tunable by strain. For compressive strain we find that the **Q**-valley shifts even further down in energy compared to the **K**-valley, suggesting an enhanced redshift of $X_{\bar{Q}}$ compared to $X_{\bar{K}}$. Applying tensile strain has the opposite effect, causing WSe_2 to behave as a direct band gap material at certain levels of strain (see SI for more details). Here, we speculate that $X_{\bar{K}}$ would dominate the low-energy spectrum⁴⁶. Quantitatively predicting the strain dependent transition between a **Q**- and a **K**-valley dominated low-energy spectrum would be an interesting avenue for future calculations.

In summary, we explored low-energy neutral and charged excitations of monolayer WSe_2 , incorporating the effects of its indirect band gap in an unprecedented manner. Our analysis reveals remarkable agreement with experimental data for excitons (**X**) and positively charged trions (X^+), along with the previously established negatively charged trions localized around the **K**-valley ($X_{\bar{K}}$). However, we identified additional bright negatively charged trions composed of configurations involving the **Q**-valley ($X_{\bar{Q}}$), exhibiting absorption characteristics similar to $X_{\bar{K}}$. Despite similarities, we observed a significant enhancement in the STS and a systematic redshift of energy in $X_{\bar{Q}}$. The increased splitting is quantitatively attributed to strong eh exchange coupling due to the greater spin polarization in the lowermost conduction band at **Q** compared to **K**.

These findings not only resonate with recent experimental and theoretical observations but also emphasize the importance of spin polarization in determining the energetic ordering of these trionic species. Our findings shine a new light on existing measurements but also has important implications for the design of future optoelectronic devices, introducing a new candidate for the field of valleytronics, harnessing the unique properties of valley-dependent phenomena.

Methods

At the core of our method is the parameter-free and truly ab initio effective many-body Hamiltonian $\mathcal{H}$ formulated in the electron-hole (eh) picture^{47,48} (see SI for more details). The approach was first implemented for semiconductor quantum dots^{49,50} where an empirical screening derived for bulk semiconductors was used. Our approach follows this original work with modifications for 2D materials as given by Torche et al.⁵¹, which includes a random phase approximation (RPA) screening. The approach naturally yields any level of charge excitation, and takes the second quantization form

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_{eh} + \mathcal{H}_{ee} + \mathcal{H}_{hh}, \quad (1)$$

where

$$\begin{aligned} \mathcal{H}_0 &= \sum_i \epsilon_i^{\text{qp}} a_i^\dagger a_i - \sum_\alpha \epsilon_\alpha^{\text{qp}} b_\alpha^\dagger b_\alpha, \\ \mathcal{H}_{eh} &= \sum_{i,j,\alpha,\beta} [\langle i\alpha|v|j\beta\rangle - \langle i\alpha|W|j\beta\rangle] a_i^\dagger b_\beta^\dagger b_\alpha a_j, \\ \mathcal{H}_{ee} &= \frac{1}{2} \sum_{i,j,k,l} \langle ij|W|kl\rangle a_i^\dagger a_j^\dagger a_l a_k, \\ \mathcal{H}_{hh} &= \frac{1}{2} \sum_{\alpha,\beta,\gamma,\lambda} \langle \alpha\beta|W|\gamma\lambda\rangle b_\lambda^\dagger b_\gamma^\dagger b_\alpha b_\beta. \end{aligned}$$

Here, a_i and $a_i^\dagger$ (b_α and $b_\alpha^\dagger$) represent electron (hole) annihilation and creation operators, respectively. The Latin (Greek) index i, j ($\alpha, \beta, \dots$) denote states above (below) the Fermi level. Moreover, $\epsilon_{i,\alpha}^{\text{qp}}$ are quasiparticle energy levels typically found from a *GW* calculation. The two-body matrix elements

are computed as

$$\langle ij|v|kl\rangle = \int \psi_i^*(1)\psi_j^*(2)v(\mathbf{r}_1, \mathbf{r}_2)\psi_k(1)\psi_l(2) d1d2,$$

$$\langle ij|W|kl\rangle = \int \psi_i^*(1)\psi_j^*(2)W(\mathbf{r}_1, \mathbf{r}_2)\psi_k(1)\psi_l(2) d1d2,$$

where $1 = (\mathbf{r}_1, \sigma)$ combines position and spin, while $\int d1 = \sum_{\sigma=\uparrow,\downarrow} \int d\mathbf{r}_1$ includes spin summation. Furthermore, ψ_i denotes single-particle wavefunctions found using density functional theory (DFT), while $v(\mathbf{r}_1, \mathbf{r}_2)$ represents the bare (unscreened) Coulomb interaction, and $W(\mathbf{r}_1, \mathbf{r}_2)$ the screened one, here computed within the RPA^{52,53}. In this work the eh exchange interaction is left unscreened, which is still debatable^{50,54}. In the SI, we compare results with and without screening of this term, similarly to Torche et al.⁵¹. Although discrepancies are minor, the unscreened results align closer with experiment.

Matrix elements for X , X^- and X^+ , can be found in refs. 51,55, and with more detail in the SI. However, our implementation uses second quantization and does not require an explicit expression of these elements.

The starting point for our many-body calculations are Kohn–Sham wave functions and energies computed within DFT using the generalized-gradient approximation and the parametrization of Perdew, Burke and Ernzerhof (PBE)⁵⁶. We utilize fully-relativistic, norm-conserving pseudopotentials⁵⁷ and employ the plane wave implementation as present in the Quantum Espresso suite^{58,59} to compute the electronic ground state observables. An out-of-plane cell length of 50 Bohr has been used to suppress the spurious interaction of periodic images. During structural relaxation, we include an ab initio DFT-D3-BJ parameterization⁶⁰ of the van-der-Waals interaction, achieving a lattice constant closer to experimental values^{61–64}. We found that a plane wave cutoff of 60 Ry yields converged results. The construction of the screened interaction kernel, W , includes the static inverse dielectric screening function, $\epsilon_{\text{GC}}^{-1}(\omega = 0, \mathbf{q})$, which we compute within the RPA using the YAMBO code^{52,53}. An energetic screening cutoff of 10 Ry, along with 20 times more unoccupied bands than occupied bands, ensuring convergence of the response function. Lastly, we use Kohn–Sham energies as IP energy levels instead of quasi-particles energies in our Hamiltonian, since our interests lie in binding energies, which depend only on two-body matrix elements. Further computational details regarding implementation and convergence can be found in the SI.

Data availability

The data used and analyzed during the current study is available from the corresponding author upon reasonable request.

Code availability

The underlying code for this study is not publicly available but may be made available to qualified researchers on reasonable request from the corresponding author.

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C.E.M.N. and F.F. carried out all calculations, analysed and visualized the results. Method development and writing were contributed equally by all authors (C.E.M.N., F.F., and G.B.)

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